Basic DAC Techniques :

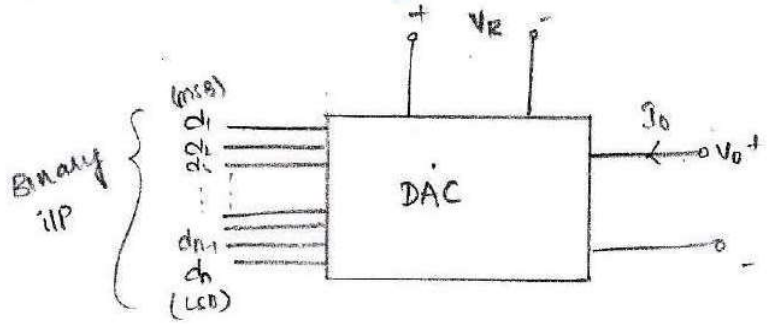


fig: Schematic of DAC

1. The i/p is an n-bit binary word D and is combined with a reference voltage V_R to give an analog o/p signal.
2. The o/p of a DAC can be either of a voltage (or) current.
3. For a voltage o/p DAC, the D/A converter is mathematically described as

$$V_o = K V_{FS} \left(d_1 z^{-1} + d_2 z^{-2} + \cdots + d_n z^{-n} \right)$$

where, V_o = o/p voltage

V_{FS} = full scale o/p voltage

K = scaling factor = unity

$d_1 \cdots d_n$ = n bit binary fractional word with the decimal point located at the left

d_1 = MSB bit

d_n = LSB bit

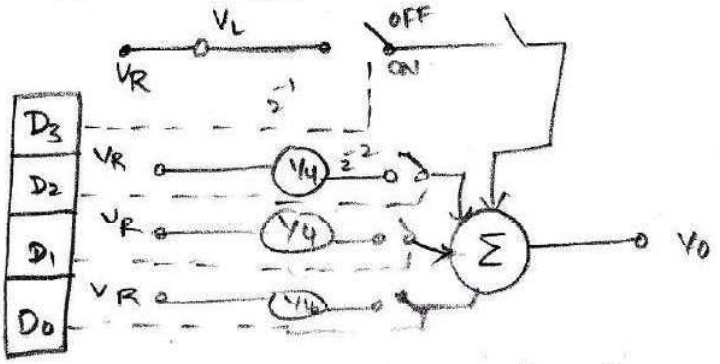


fig. Digital to Analog Conversion

→ Fig. shows explanation of DAC. Here V_R is fixed reference voltage.

→ The digital signal is contained in the register on the left.

→ Each flipflop in the register controls an analog switch as shown in the dotted lines.

$$V_0 = \left[D_3 2^{-1} + D_2 2^{-2} + D_1 2^{-3} + D_0 2^{-4} \right] V_R$$

If $D_3\ D_2\ D_1\ D_0 = 1001$

$$V_0 = 1\left[\frac{V_R}{2} + 0\,\frac{V_R}{4} + 0\,\frac{V_R}{8} + 1\,\frac{V_R}{16} \right]$$

$$\boxed{V_0 = \frac{9V_R}{16}}$$

→ There are 2 types D/A converters

1. Weighted - resistor D/A converter
2. R-2R ladder D/A converter.

1. Weighted - Resistor D/A Converter :

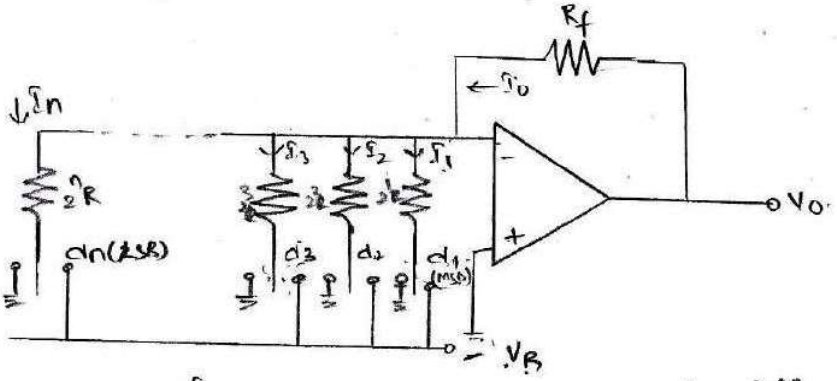


fig: A simple weighted Resistor DAC

→ One of the simplest CKT uses a summing amplifier with a binary weighted resistor NTWK. It has n-electronic switches $d_1, d_2 \cdots d_n$ controlled by binary i/p word.

→ These switches are SPDT (single pole double throw) types switch

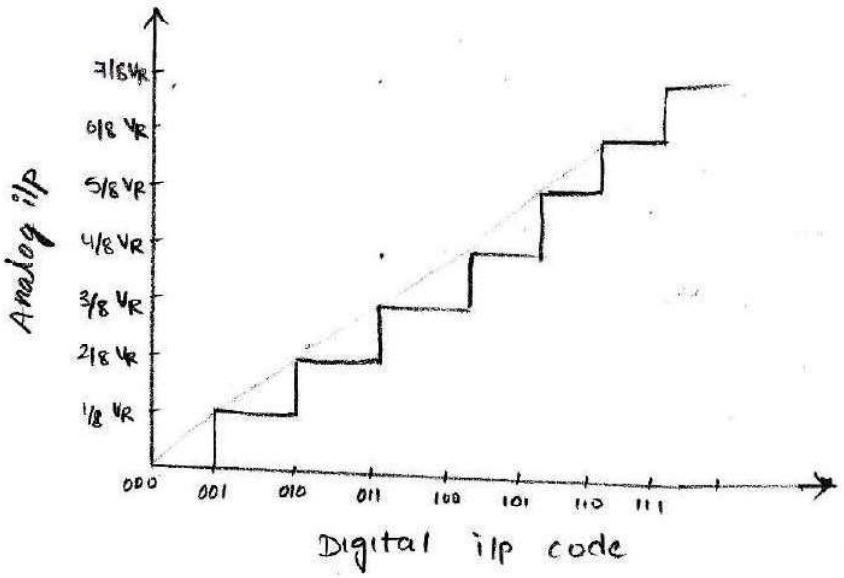


fig: Transfer Char'n of a 3 bit DAC

Eg: If the binary i/p to a particular switch is 1, its connets the resistance to the reference voltage ($-V_R$).

→ The binary weighted resistor DAC uses an op-amp to sum of 'n' binary weighted currents derived from a reference voltage V_R via current scalling resistors 2R, 4R, 8R ... 2^nR as shown in the fig.

Working:

In fig. switch positions are controlled by the digital i/p's. When digital i/p logic 1, it connects the corresponding resistance to the reference voltage V_R otherwise it leaves resistor open.

∴ for ON switch $I = \frac{-V_R}{R}$

for OFF Switch $I = 0$

Here op-amp is used as a summing amplifier. Due to high i/p impedence of op-amp, summing current will flow through R_f. Hence the total current through R_f can be given as.

$$I_0 = I_1 + I_2 + I_3 - \cdots - I_n$$

The o/p voltage is the voltage across R_f and it is given as

$$\text{so} \quad V_0 = -I_0 R_f$$

$$V_0 = -\left[I_1 + I_2 + I_3 + \cdots I_n\right] R_f$$

$$V_0 = \left[D_1 \frac{V_R}{2R} + D_2 \frac{V_R}{4R} + D_3 \frac{V_R}{8R} + \cdots D_n \frac{V_R}{2^n R}\right] V_f$$

$$= \frac{V_R}{R} R_f \left[D_1 2^{-1} + D_2 2^{-2} + \cdots D_n 2^{-n}\right]$$

When $R_f = R$

$$\therefore V_0 = V_R \left[D_1 2^{-1} + D_2 2^{-2} + D_3 2^{-3} + \cdots D_n 2^{-n}\right]$$

→ The above eqn indicates that the analog o/p voltage is proportional to the i/p digital word.

Drawbacks :

1. Wide range of resistor values are required.

For 8 bit DAC, the resistor required are 2^1R, 2^2R, 2^3R ... and 2^8R. Therefore, the largest resistor is 128 times the smallest one.

This wide range of resistor values has restrics on both higher & lower ends.

The finite resistance of the switches disturbs the binary weighted relationship among the various currents, particularly in the MSB positions, where the current setting resistances are smaller.

2. R-2R Ladder DAC :

→ In this type, reference voltage is applied to one of the switch positions and other switch position is connected to ground. It is shown in below figure.

→ Wide range of resistors are required in binary weighted resister type DAC. This can be avoided by using R-2R ladder type DAC where only 2 values of resistors are required.

→ The typical value of R ranges from 2.5 KΩ to 10 KΩ.

→ For simplicity let us consider a 3-bit DAC. where the switch position $D_1\ D_2\ D_3$ corresponds to the binary word 100.

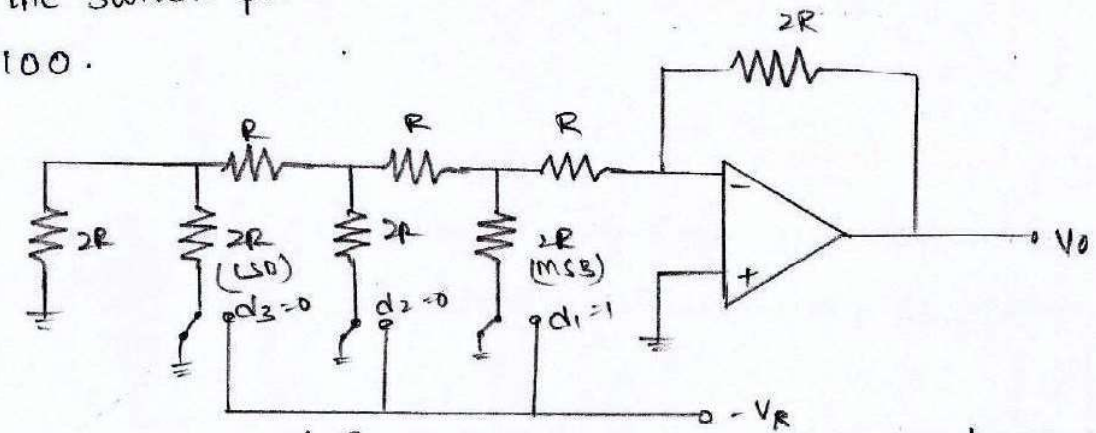


fig ⓐ

→ The equivalent circuit can be simplified as shown in below fig.

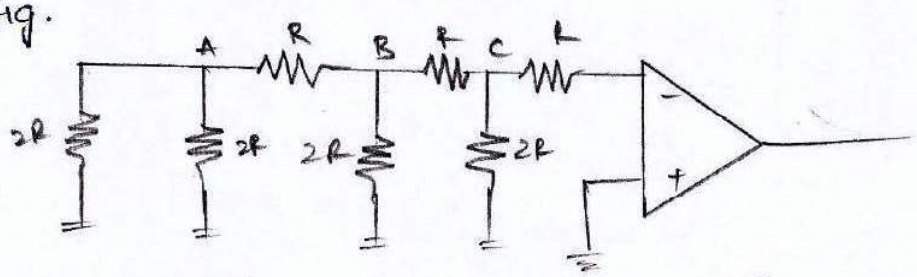
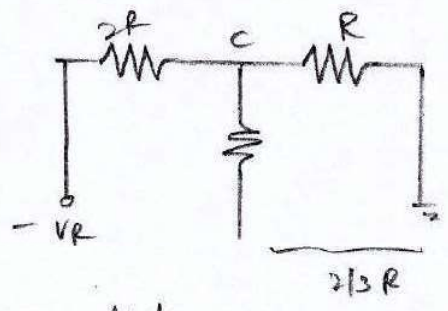


fig ⓑ Equivalent ckt of fig ⓐ



equivalent ckt of fig ⓑ

→ Then voltage at node c can be easily calculated by the set procedure of network analysis as

$$\frac{-V_R\left(\frac{2}{3}R\right)}{2R+\frac{2}{3}R} = \frac{-V_R}{4}$$

The o/p voltage is

$$V_o = -\frac{2R}{R}\left(\frac{-V_R}{4}\right)$$

$$= \frac{V_R}{2}$$

$$\boxed{V_o = \frac{V_{FS}}{2}}$$

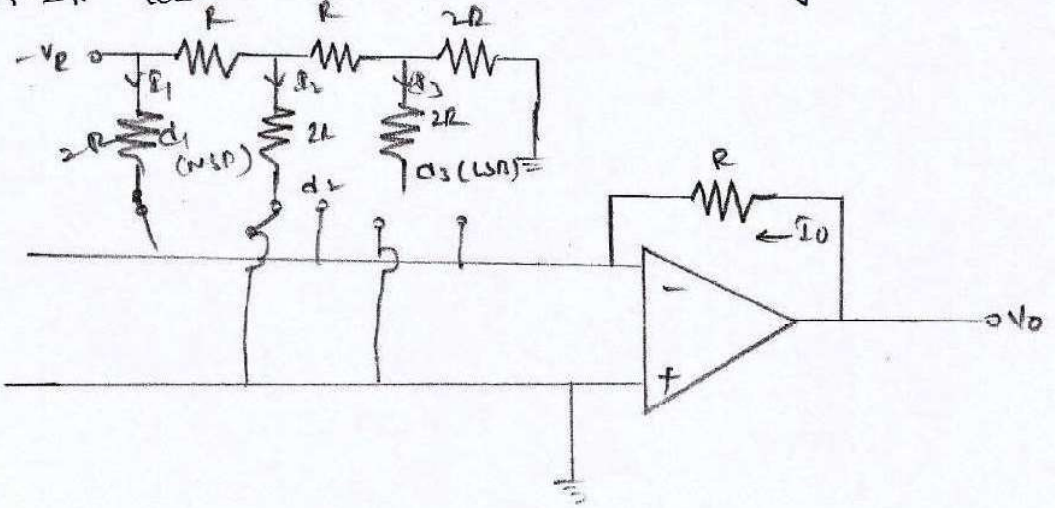
3. Inverted R-2R ladder :

→ In weighted resistor type DAC and R-2R ladder type DAC current flowing in the resistors changes as the i/p date changes.

→ more power dissipation causes heating.

→ this problem can be avoided completely in inverted R-2R ladder type DAC.

→ R-2R ladder D/A convertor uses only 2 resistors values.



→ Here each bit of binary word converts the corresponding switch either to ground (or) to the inverting i/p terminal of the op-amp which is at the virtual ground.

→ Since both the positions of switch are at ground potential, the current flowing through resistances is constant and it is independent of switch position.

→ These current can be given as

$$I_1 = \frac{V_R}{R}$$

$$I_2 = \frac{V_R/2}{2R}$$

$$= \frac{V_R}{4R}$$

$$I_2 = \frac{I_1}{2}$$

$$I_3 = \frac{V_R/4}{2R}$$

$$= \frac{V_R}{8R}$$

$$I_3 = \frac{I_1}{4}$$

We know that,

$$V_0 = -I_0 R_f$$

$$= -R_f (I_1 + I_2 + I_3)$$

$$V_0 = -R_f \left[D_1 \frac{V_R}{2R} + D_2 \frac{V_R}{4R} + D_3 \frac{V_R}{8R} \right]$$

$$= -\frac{V_R R_f}{R} \left[D_1 2^{-1} + D_2 2^{-2} + D_3 2^{-3} \right]$$

If $R_f = R = V_0 = -V_R \left(D_1 2^{-1} + D_2 2^{-2} + D_3 2^{-3} \right)$

Let us consider 3-bit i/p with binary 011.

Here the o/p voltage is given as

$$V_O = -V_R \left[0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}\right]$$

$$= -V_R \left[0 + \frac{1}{4} + \frac{1}{8}\right]$$

$$= -V_R \left[0.25 + 0.125\right]$$

$$= -V_R \left[0.375\right]$$

Let $V_R = 5\,V$

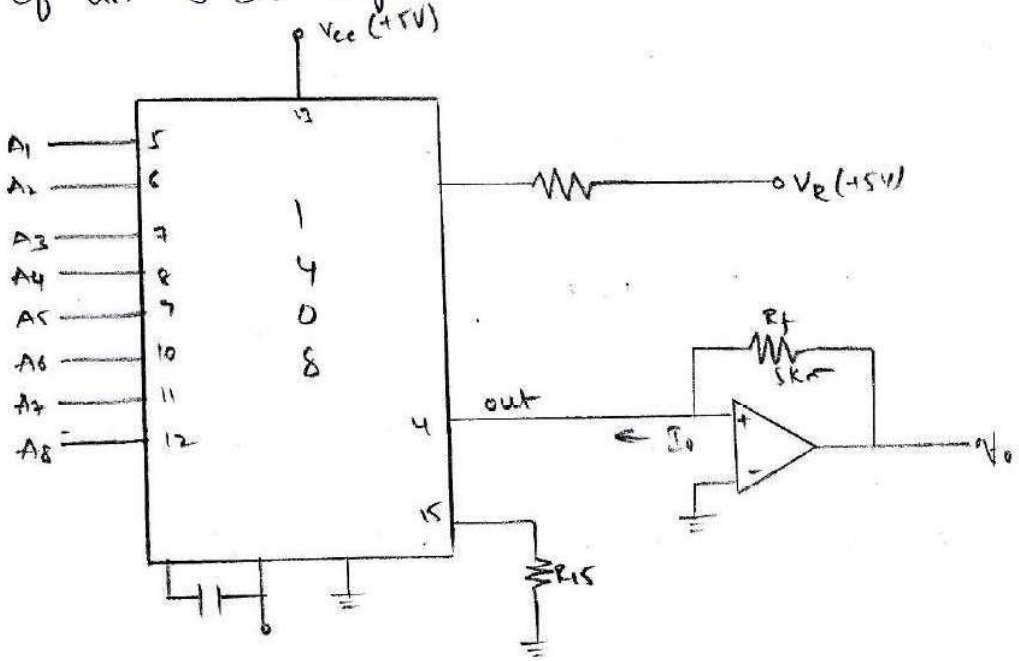
$$= -5 \times 0.375$$

$$\boxed{V_O = -1.875\ V}$$

IC 1408 DAC :

→ The 1408 is an 8-bit R-2R ladder type D/A convertor compatible with TTL and CMOS logic.

→ It is designed to use where the o/p current is linear product of an 8-bit digital word.



→ The IC 1408 consists of a reference current amplifier, an R/2R ladder and 8-bit high speed current switch.

→ It has 8 i/p data lines as A1 (MSB) through A8 (LSB) which controls the positions of current switches.

→ It requires 2mA reference current for full scale i/p & 2 power supplies $V_{CC} = +5V$ and $V_{EE} = -5V$.

→ The voltage V_R and resistor R_{14} determines the total reference current source.

$$\text{i.e., Current } I = \frac{V_R}{R_{14}}$$

$$= \frac{5}{2.5k\Omega} = 2mA.$$

→ R_{15} is generally equal to R_{14} to match the i/p impedence of the current amplifier.

→ The below fig. shows a typical 8-bit DAC 1408 compatible with TTL and CMOS logic.

→ If settling time is around 300 ns.

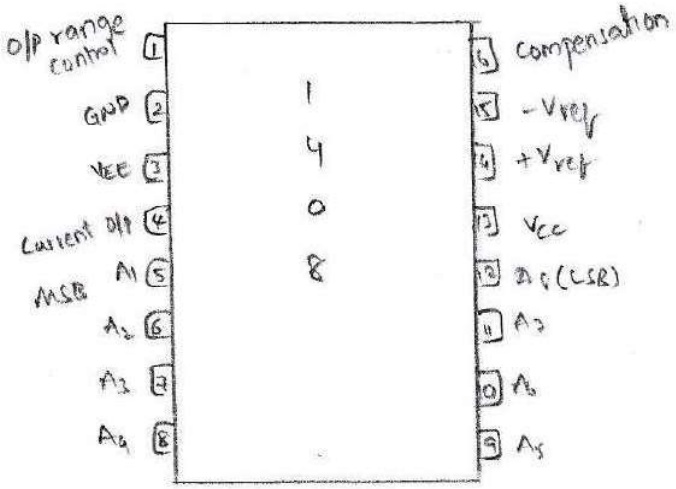


fig : pin diagram

→ It consist of 8 i/p data lines A_1 (MSB) to A_8 (LSB).

→ It needs 2mA reference current for full scale i/p.

→ The two power supplies $V_{CC} = 5V$ & $V_{EE} = -5V$.

→ The resistance R_{15} is equal to R_{14}. Hence match the i/p impedence of the reference source.

The o/p current I_o is given by

$$I_o = \frac{V_R}{R_{14}} \left[\sum_{i=1}^{n} A_i 2^{-i} \right]$$

$A_i = 0$ (or) 1

The total reference current is equal to

$$\frac{V_R}{R_{14}} = 5V / 2.5K\Omega = 2mA.$$

for full scale i/p

$$I_o = \frac{5}{2.5K} \left[\sum_{i=1}^{8} A_i \times 2^{-i} \right]$$

$$= 1.992\ mA$$

The o/p voltage V_o for full scale i/p is

$$V_o = 2mA \left[\frac{255}{256} \right] \times 5 \times 10^3$$

$$= 9.91\ V.$$

The o/p voltage equ

$$V_o = \frac{V_R}{R_{14}} R_f \left[\frac{d_1}{2} + \frac{d_2}{4} + \frac{d_3}{8} + \cdots + \frac{d_8}{256} \right]$$

→ for the bipolar range from -5V to 5V the 1408 DAC can calibrated by adding R_B blw V_R and o/p voltage pin 4

as indicator in the following fig.

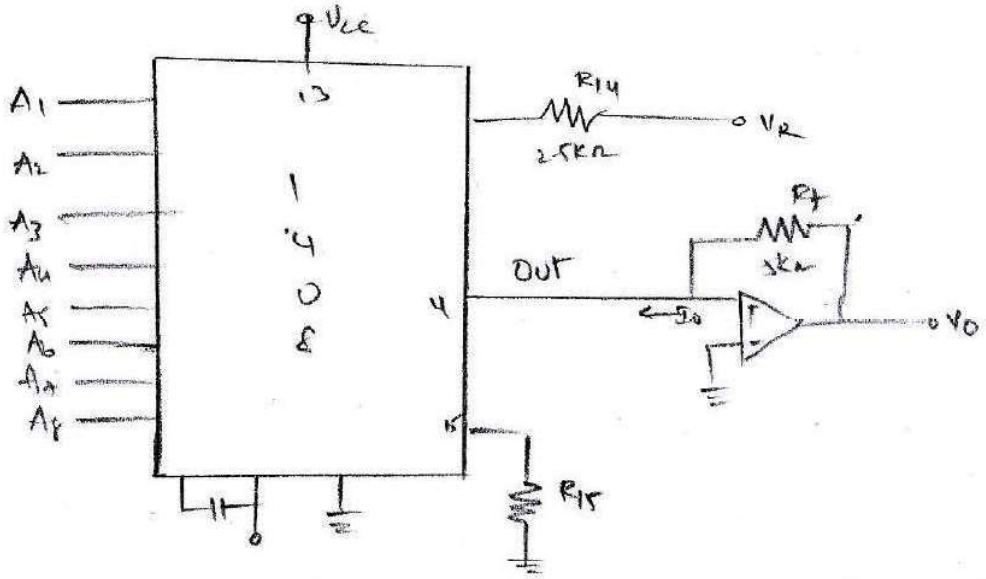
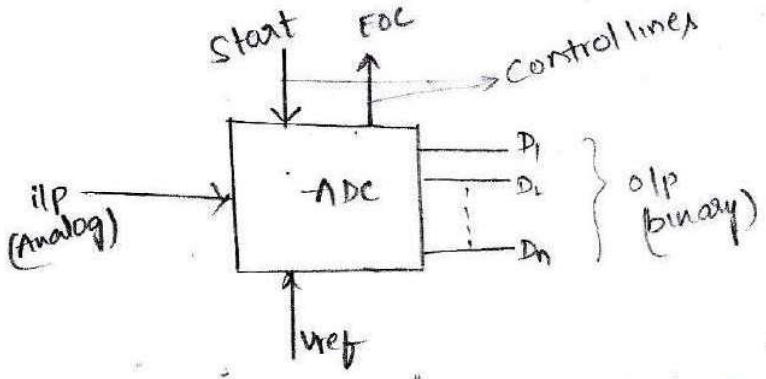


fig: 1408 D/A converter voltage o/p range in unipolar Range

→ The resistor R_B gives 1mA current to the o/p. The o/p current for the bipolar operation I_o' is

$$I_o = I_o - \left[\frac{V_R}{R_{14}}\right]\left[\sum_{i=1}^{8} d_i 2^{-i}\right] - \left(\frac{V_B}{R_B}\right)$$

Analog to Digital Convertors :



EOC = end of conversion

→ The ADC, is a circuit in which analog i/p combines with reference voltage to give the digital o/p. The schematic representation of ADC is shown in above fig.

→ ADC is a opposite version of DAC. ADC also uses the reference voltage to give an O/P.

The o/p eqn is

$$D = d_1 2^{-1} + d_2 2^{-2} + \cdots d_n 2^{-n}.$$

where d_1 = MSB ; d_n = LSB

→ ADC consists of 2 control lines i.e., start & EOC.

Start : It is a i/p control line to tell the ADC to start the conversion.

EOC : It is a o/p control line to announce when the conversion is completed.

→ Depending upon the type of application ADC are designed for micro processor.

→ ADC are classified into 2 types. 1. Direct type ADC
2. Integrating type ADC.

1. Direct type ADC :

It compares the given analog signal with the internally generated equivalent signal. There are classified into 3 types.

(a) Parallel Comparator type ADC

(b) Counter type ADC

(c) Successive Approximation type ADC.

2. Integrating Type ADC :

It performs conversion in indirect manner by first changing the analog i/p signal to a linear function of time & then to a digital bits. There are classified into 2 types.

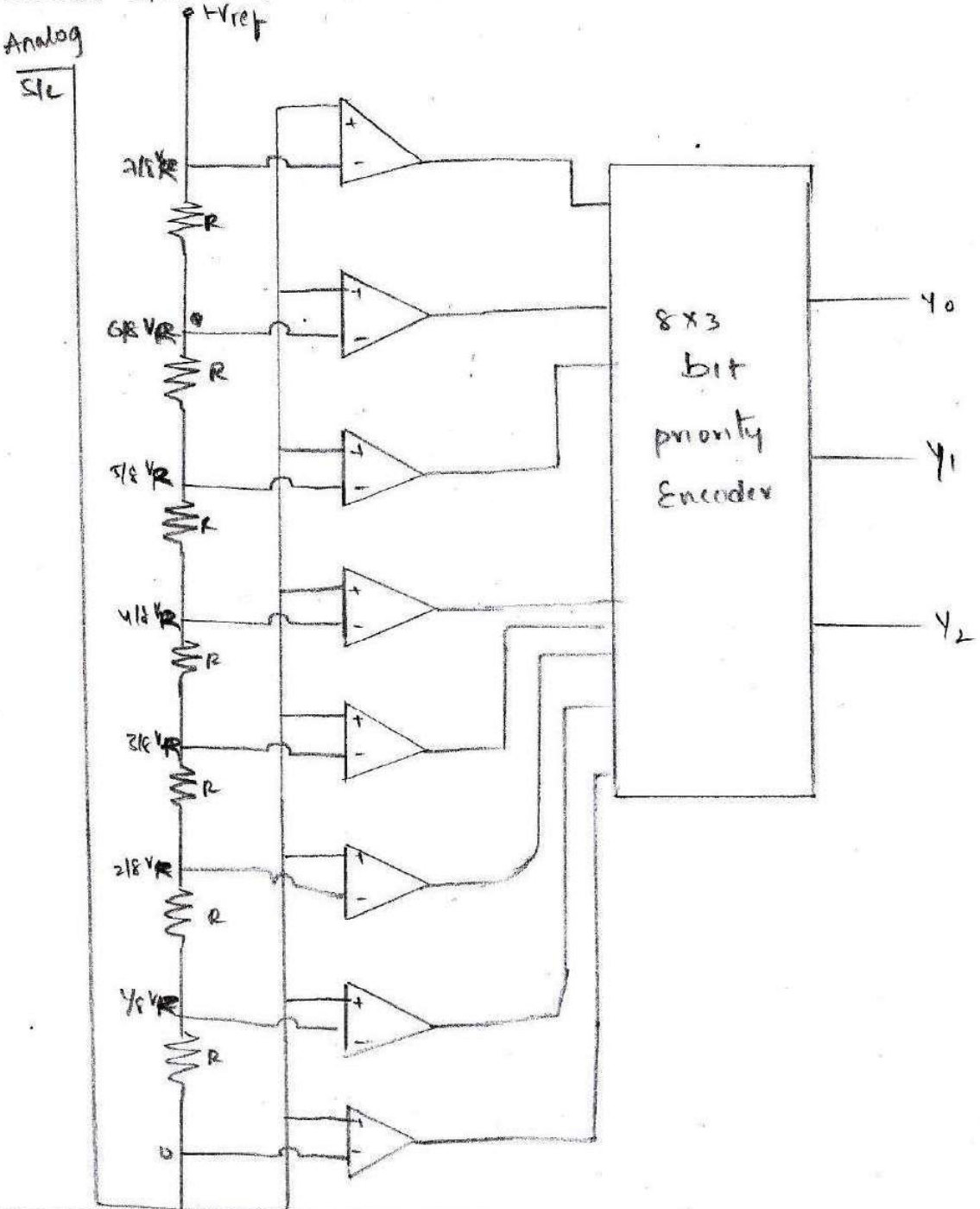
(a) Dual slope type ADC

(b) Charge Balance type ADC.

(a) Parallel Comparator type ADC :

It is the simplest convertor at the same time it is fastest & most expensive technique.

→ The fig. shows the 3-bit A/D convertor. It consist of resistive divider n/w. Op-Amp comparators & priority encoder.



Va	I0	I1	I2	I3	I4	I5	I6	I7	Y0	Y1	Y2	Y
0 to 1/8 VR	1	0	0	0	0	0	0	0	0	0	0	
1/8 to 2/8 VR	1	1	0	0	0	0	0	0	0	0	1	

2/8 V_R to 3/8 V_R	1	1	1	0	0	0	0	0	1	0
3/8 to 4/8	1	1	1	1	0	0	0	0	1	1
4/8 to 5/8	1	1	1	1	1	0	0	1	0	0
5/8 to 6/8	1	1	1	1	1	1	0	1	0	1
6/8 to 7/8	1	1	1	1	1	1	1	1	1	0

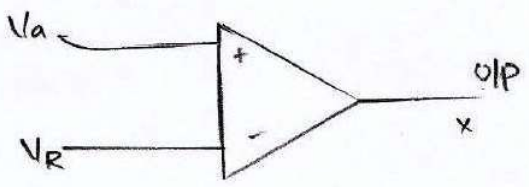
→ One i/p of each comparator is connected to the i/p signal Va & other i/p is connected to the reference signal V_R generated by the resistive divider nlw.

→ At each node of the resistive divider equal voltage is available.

→ Since all the resistors are of equal value, the voltage levels available at the nodes are equally divided blw V_R & GND.

→ The purpose of the circuit is to compare the analog i/p Va with each of the node voltages. Corresponding truth table is shown in above fig.

→ The individual comparator truth table is shown in below fig.



i/p's	o/p
$Va > V_R$	$x = \phi$
$Va < V_R$	$x = 0$
$Va = V_R$	previous o/p

→ It is also called as Flash type A/D comparator.

(b) Counter type ADC :

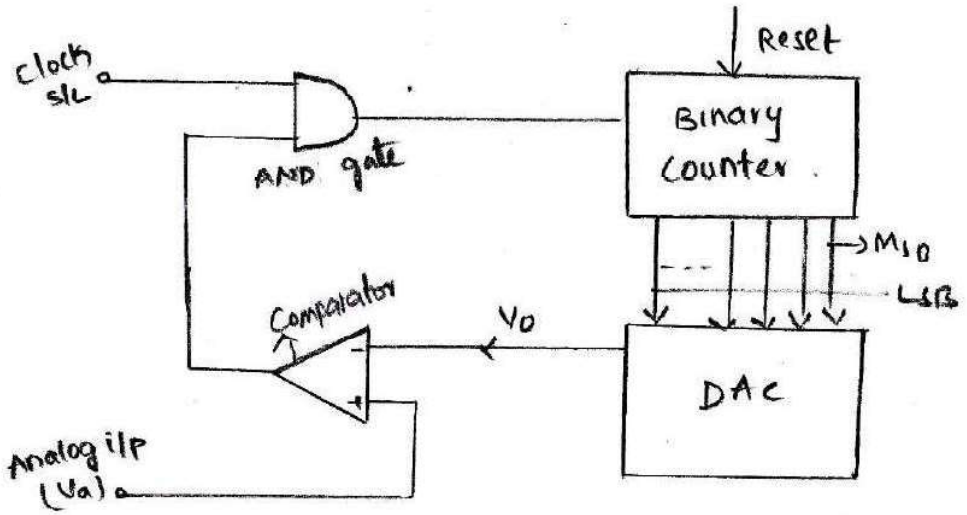
It is one type of analog to Digital convertor.

→ Here the o/p of DAC is continously compared with the i/p analog signal.

→ When the o/p of DAC is greater than analog i/p then only the o/p of comparator is high.

→ The block diagram of counter type ADC is shown in fig.

→ It consists of a binary counter, DAC, comparator & AND gate.



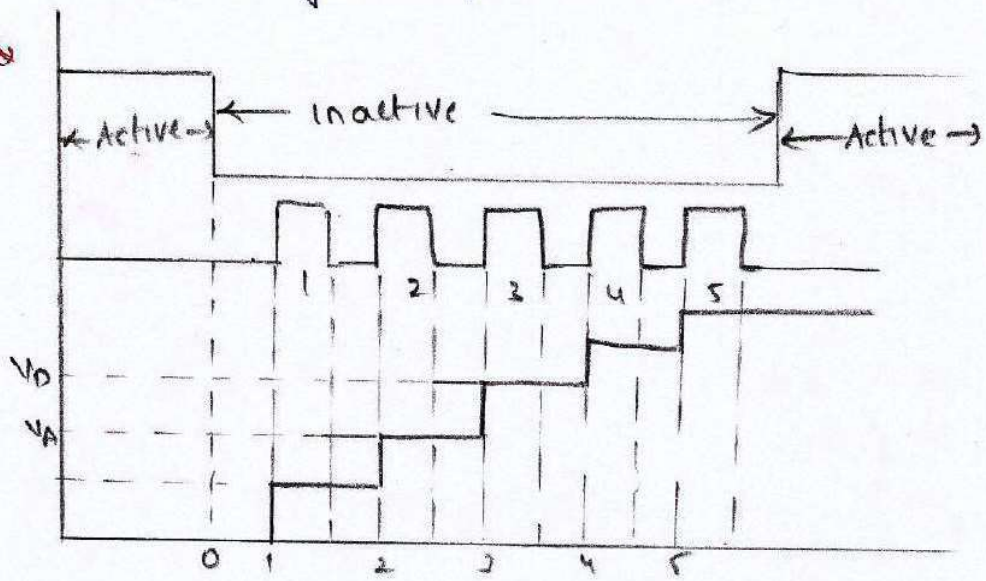
Working :

1. Initially the counter is reset ie., its o/p is set to zero by applying a reset pulse.
2. The o/p of counter is given as digital i/p to DAC. Since i/p to DAC is zero. Its o/p is zero.
3. When the analog i/p is applied to comparator, it becomes greater than V_D.
4. Here V_A is applied to the non-inverting terminal of the comparator & V_D is applied to the inverting terminal of the comparator. When $V_A > V_D$, the comparator o/p goes high.
5. For an "AND" gate one i/p is clock pulse applied & another i/p is the o/p of the comparator.
6. When comparator o/p is high then the AND gate is able to allow the clock pulses & it is given to the binary counter.
7. The counter starts counting these clock pulses according to the clock pulses goes on increasing. This increases the o/p of the DAC.

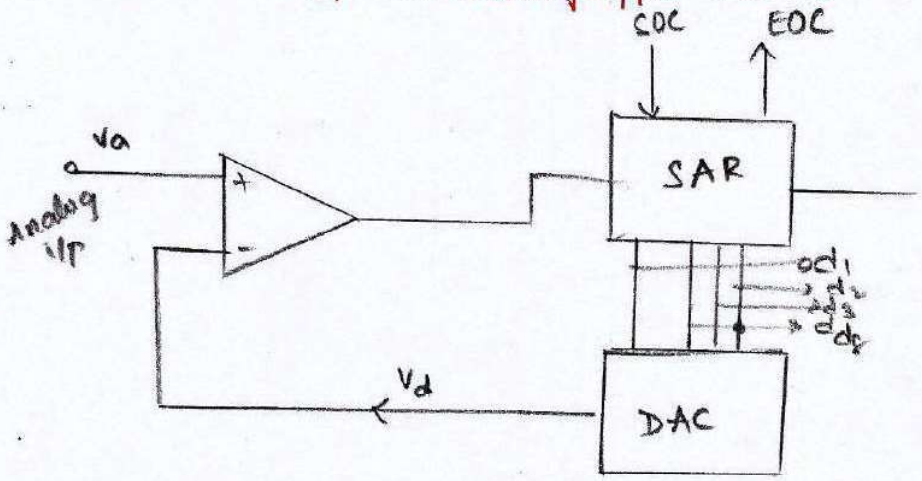
The above steps are repeated till $V_A < V_D$. when V_A is less than V_D the o/p of comparator goes low due to this 'and' gate disables i.e., AND gate is not allow the clock pulses so the counting process of counter is stopped.

→ For next A/D conversion the i/p voltage V_n changes. The binary counter is cleared by applying second reset pulse.

→ The corresponding waveforms shown in below figure.



(c) Successive Approximating type ADC :



→ It is one of the most popular A/D convertor. The block diagram of successive approximation type ADC shown in above figure.

→ It consists of SAR (Successive Approximation register), SOC, EOC and DAC, comparator.

→ The successive approximation method uses ~~process~~ for to search binary bits, it completes the searching process for n-bit conversion in 'n' clock ~~pl~~ cycles.

→ The external clock signal is used to set the internal timing parameters.

→ The control signal SOC is used to start the conversion and EOC is used to end the conversion.

→ The analog i/p signal V_a is applied at one i/p of the comparator before slot the SOC the o/p should be zero.

→ After initializing the SOC, the SAR sets the four bit binary code as an input of DAC. i.e., its sets the MSB bit $d_1 = 1$, while all other bits to zeros. So that the trail code is 1000.

→ Now the o/p of DAC is compared with the analog i/p signal V_a.

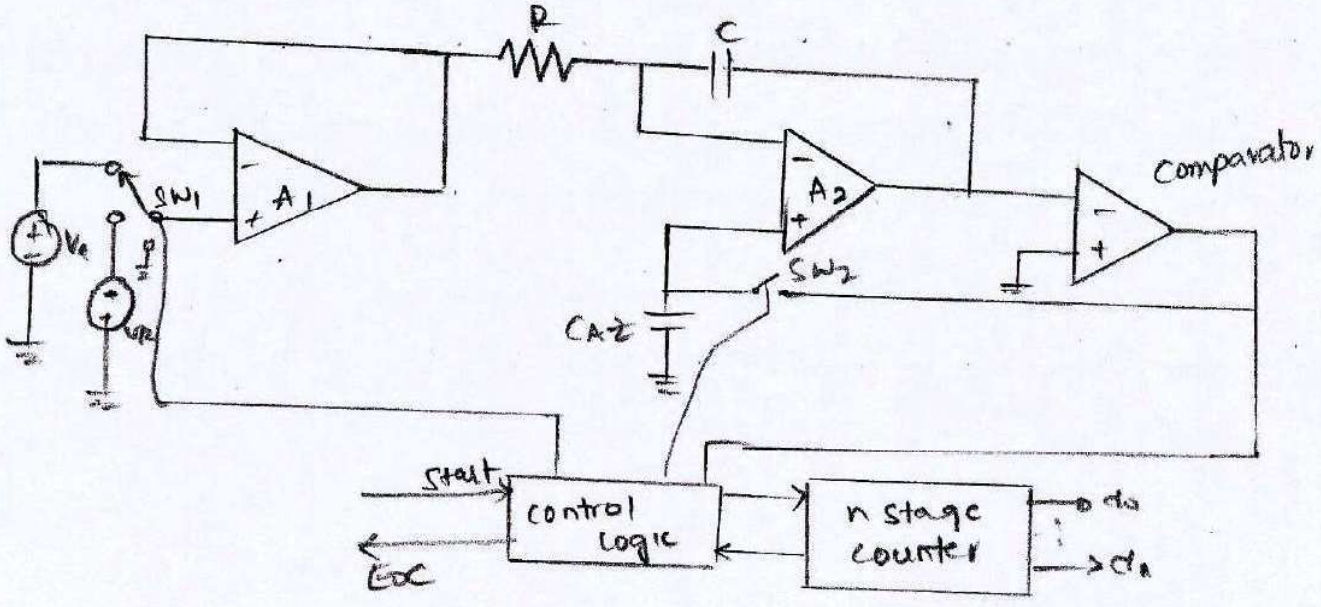
→ If $V_a > V_d$, the o/p of comparator is one then the MSB bit (or) SCR sets the MSB bit is left 1 and next LSB bit is made one and further tested.

→ However if $V_a < V_d$, the o/p of comparator is zero then SCR sets the MSB bit zero and next LSB bit is made one.

→ This process repeated for all subsequent bits, one at a time until all bit positions have been tested.

→ Whenever the DAC o/p crosses V_A, the comparator changes the state and this can be taken as the EOC control signal.

Dual Slope Type ADC :



→ The above figure shows the dual slope type ADC.

→ It consists of high input impedence buffer A_1, integrator A_2 and voltage comparator, control logic & n-stage counter.

→ The convertor first integrates the analog i/p signal V_a for a fixed duration of 2^n clock periods.

→ Then it integrates an reference voltage V_R of opposite polarity until the integrator o/p is zero.

→ The number of N ~~of~~ clock cycles required to return the integrated to zero is proportional to the value of V_a average over the integration period.

→ Before the start signal switch sw_1 is connected to ground and switch sw_2 is closed.

→ So, any ~~effect~~ offset voltage present in the A_1 & A_2, it appears across the capacitor in C_{A2} till the threshold of the comparator is achieved. Thus the capacitor C_{A2} is provides the input offset voltage of all the 3 amplifiers.

→ Later when sw_2 opens C_{A2} acts as memory device to store the energy levels (or) voltage.

→ After START signal initiates ~~the~~ at $t = t_1$, the control logic open sw_2 switch and sw_1 switch connects to the V_A and enables the counter starting from zero.

→ Here n stage ripple counter is used so the counter resets to zero after counting 2^n clock cycles (or) pulses.

→ If the clock period is t_1, the integration takes place for a time $t_1 = 2^n T$ and the o/p is a ramp going downward shown in below figure.

→ The counter resets itself to zero at the end of the interval T and the switch Sw_1 is connected to the reference voltage $-V_R$.

→ Now, the o/p voltage V_o will have positive slope.

→ However when V_o becomes just zero at a time $t = t_3$ the control logic issues an end of conversion signal. So no further clock pulses enter the counter.

→ It can be shown that the reading of the counter at t_3 is proportional to the analog i/p voltage V_a.

$$T = t_2 - t_1$$

$$= 2^n \cdot \frac{\text{clock period}}{\text{clock pulses}}$$

$$t_3 - t_2 = n \frac{\text{clock cycles}}{\text{clock period}}$$

for an integrator

$$\frac{\Delta V_o}{\Delta t} = \frac{-1}{RC} \Delta V_t$$

$$\Delta V_o = -\frac{1}{RC} \Delta V_t$$

The voltages V_o will be equal to V_1 at the instant t_2 and it can be written as

$$V_1 = -\frac{1}{RC} V_a (t_2 - t_1)$$

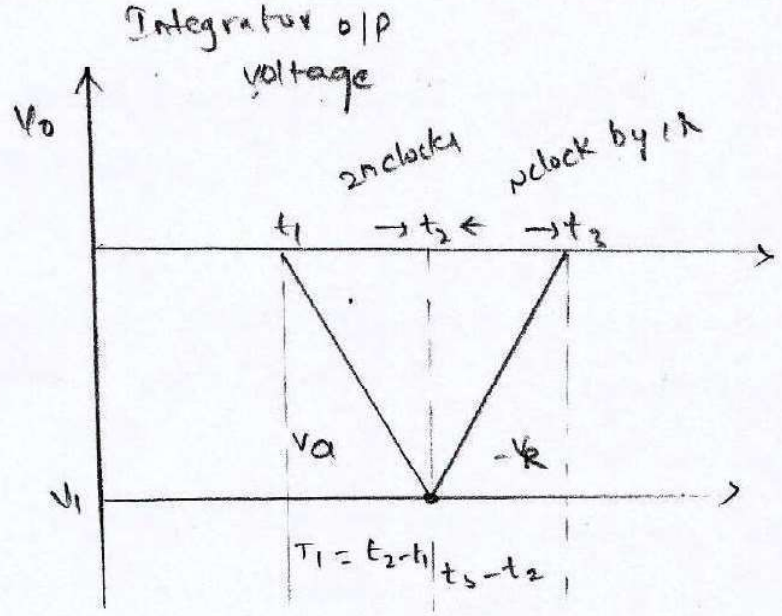
It can be also written as

$$V_1 = \frac{-1}{RC} V_R (t_3 - t_2)$$

$$V_a (t_2 - t_1) = -V_R (t_3 - t_2)$$

$$V_a (2^n) = -V_R (N)$$

$$V_a = \frac{-V_R (N)}{2^n}$$



Specifications of A/D and D/A convertor :

1. Resolution
2. Linearity
3. Accuracy
4. Settling time
5. Stability

Both A/D & D/A convertor are available with wide range of specifications.

→ The various important specifications of convertor generally specified by the manufacture are analysing.

Resolution :

The resolution of a convertor is the smallest change in voltage which may be produced at the o/p of the convertor.

Linearity :

Linearity specification of convertors. It is an important measure of accuracy and tells us how close the convertor o/p is to its ideal transfer characteristics.